

Lok Sabha Election 2019 – Data Analysis Project

Statistical Analysis | Tableau
Visualizations | BigQuery SQL

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Objective & Dataset Overview

Objectives of the Study :

- To analyze **Lok Sabha Election 2019** results for **five selected states**.
- To examine **party-wise vote share** using pie charts.
- To compare the **performance of winning and losing candidates** in each state.
- To compare **voting percentages of winning candidates** across all five states.
- To present insights using **Tableau visualizations** and **SQL analysis in BigQuery**.

Dataset Overview

- Election Year: 2019
- Datasets Used:
 - all_states_ge.csv** – Candidate-wise election data
 - ge_parties** – Party details
- Key Fields:
 - State Name
 - Candidate Name
 - Gender
 - Party Name
 - Votes Received
 - Vote Percentage
 - Margin Percentage

Tools Used

- Google Sheets – Data preparation & charts
- Tableau – Visual analytics
- Google BigQuery – SQL queries & JOIN operations

States Selected for Study

The following five states were selected to analyze the **Lok Sabha Election 2019** results:

- **Maharashtra**
- **Uttar Pradesh**
- **West Bengal**
- **Tamil Nadu**
- **Bihar**

Scope of Analysis

- Party-wise vote percentage for each selected state.
- Comparison of **winning and losing candidates** within each state.
- Cross-state comparison of **winning candidates' voting percentages**.

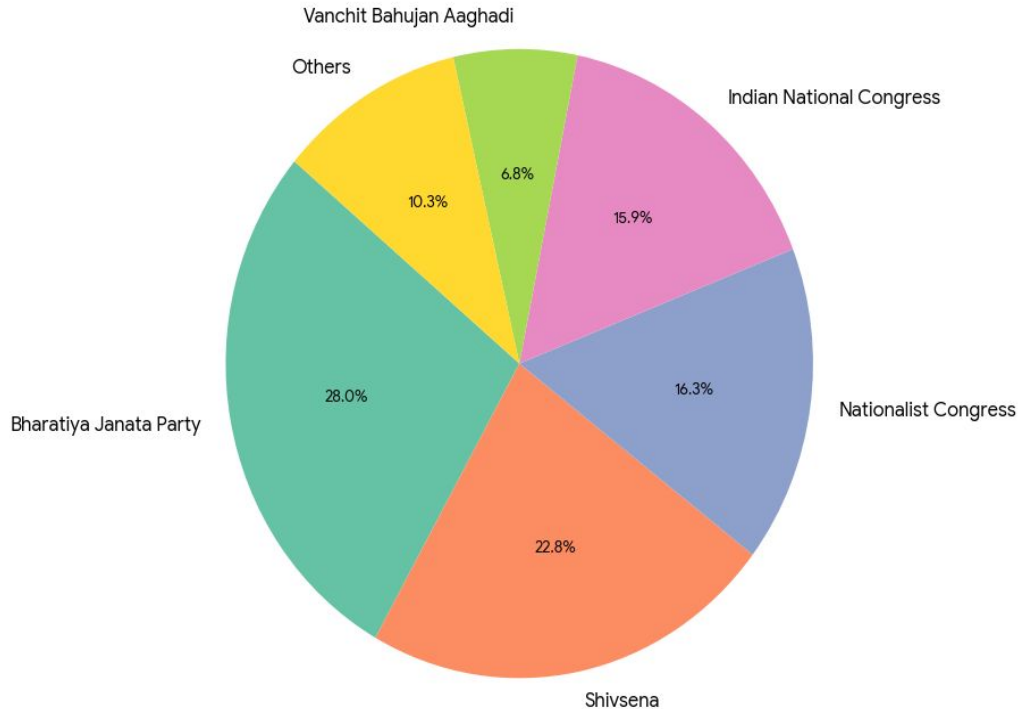
Reason for State Selection

- These states represent **different geographical regions** of India.
- They have **significant voter participation** and political importance.
- Include both **national and strong regional parties**.
- Help in understanding **diverse voting behavior and trends** across states.

Party-wise Vote Share – Maharashtra (Lok Sabha 2019)

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Vote Share by Party: Maharashtra (2019)



Objective :

To understand the **distribution of votes among political parties** in ***Maharashtra*** during the **2019 Lok Sabha election**.

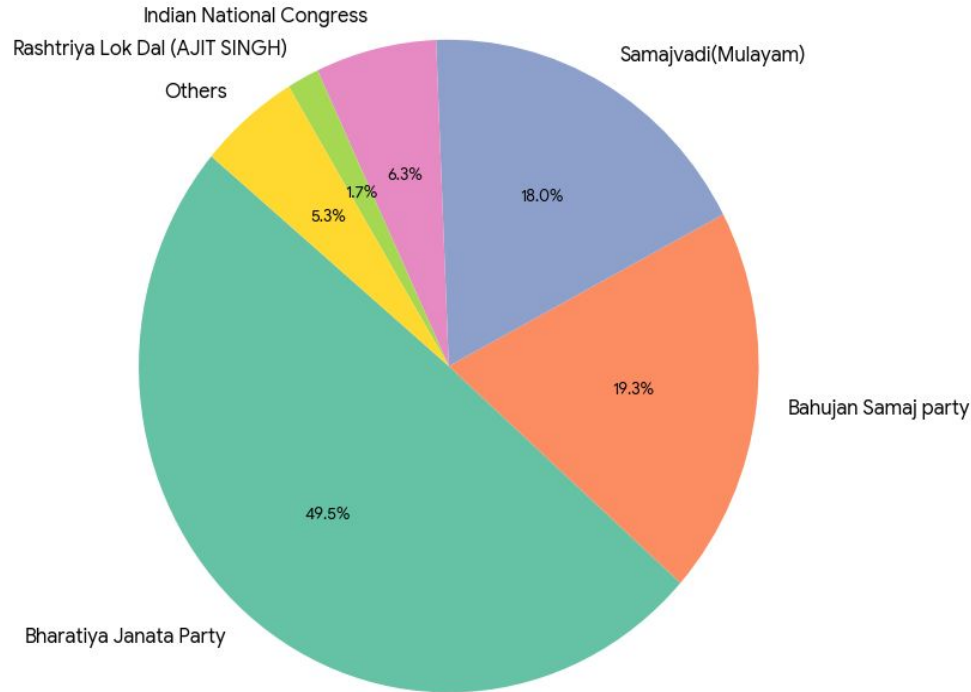
Observation :

- The **BJP** secured the **highest share of votes** in the state.
- Major national and regional parties together account for a **significant portion of total votes**.
- Smaller parties contribute a **limited share**, indicating concentrated voter preference.

Party-wise Vote Share – Uttar_Pradesh(Lok Sabha 2019)

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Vote Share by Party: Uttar_Pradesh (2019)



Objective :

To understand the **distribution of votes among political parties** in **Uttar Pradesh** during the **2019 Lok Sabha election**.

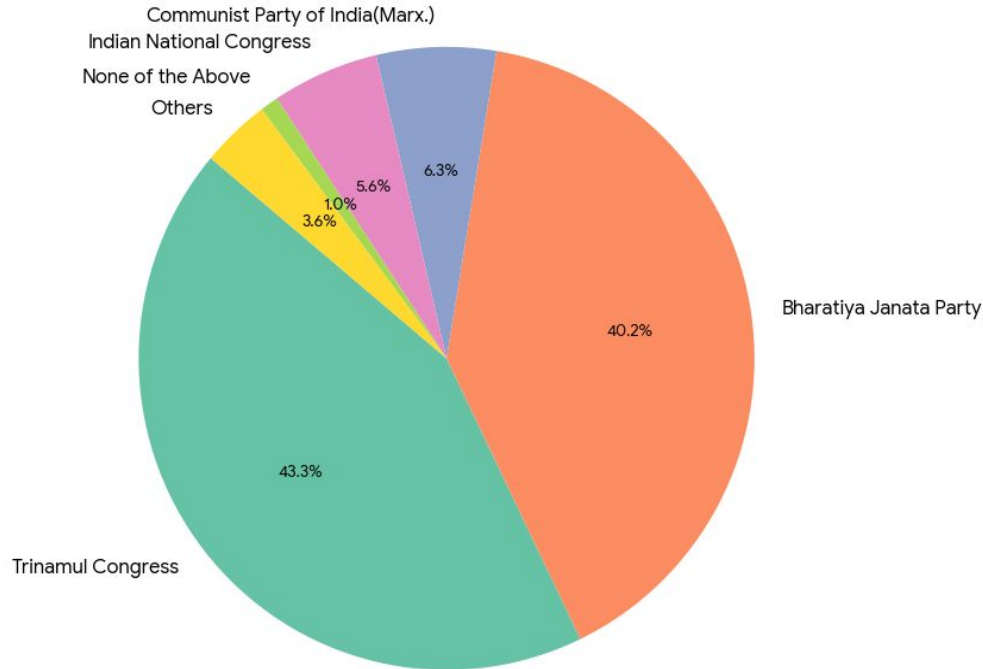
Inference :

- Voting in the state shows **dominance of major parties**.
- The vote share distribution reflects **clear electoral trends** and voter alignment.

Party-wise Vote Share – West Bengal (Lok Sabha 2019)

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Vote Share by Party: West_Bengal (2019)



Objective :

To understand the **distribution of votes among political parties in *West Bengal*** during the **2019 Lok Sabha election**.

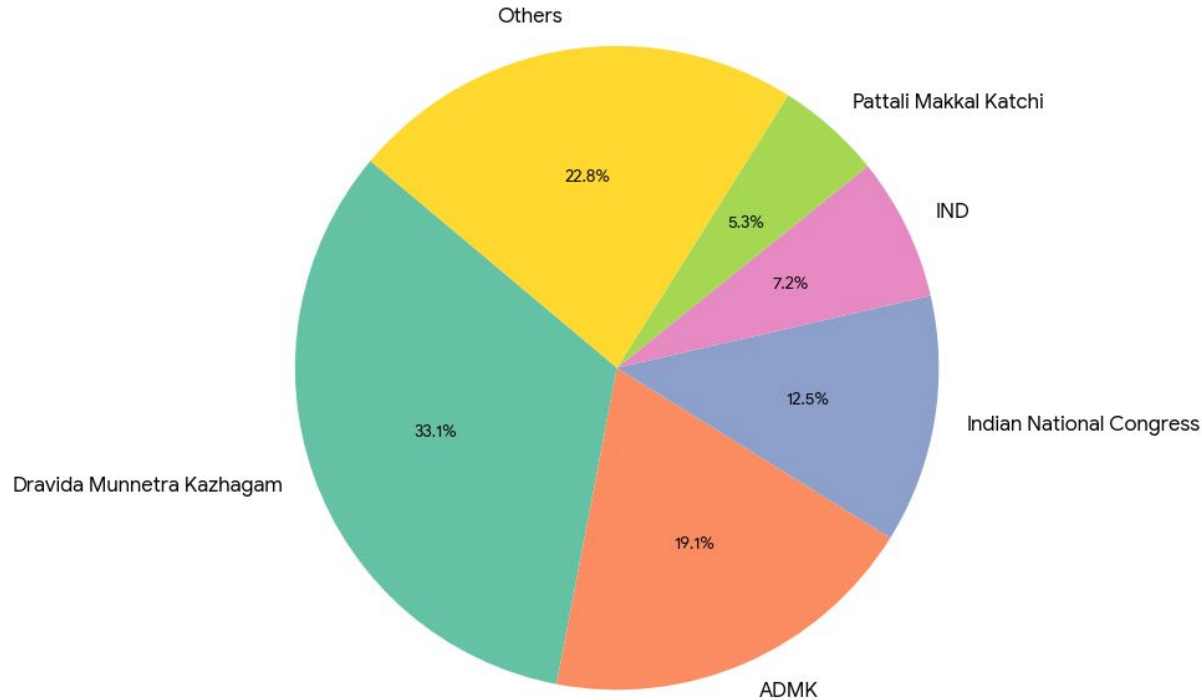
Observation :

- The leading party secured the **highest share of votes** in the state.
- Major national and regional parties together account for a **significant portion of total votes**.
- Smaller parties contribute a **limited share**, indicating concentrated voter preference.

Party-wise Vote Share –Tamil Nadu(Lok Sabha 2019)

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Vote Share by Party: Tamil_Nadu (2019)



Objective :

To understand the **distribution of votes among political parties** in **Tamil Nadu** during the **2019 Lok Sabha election**.

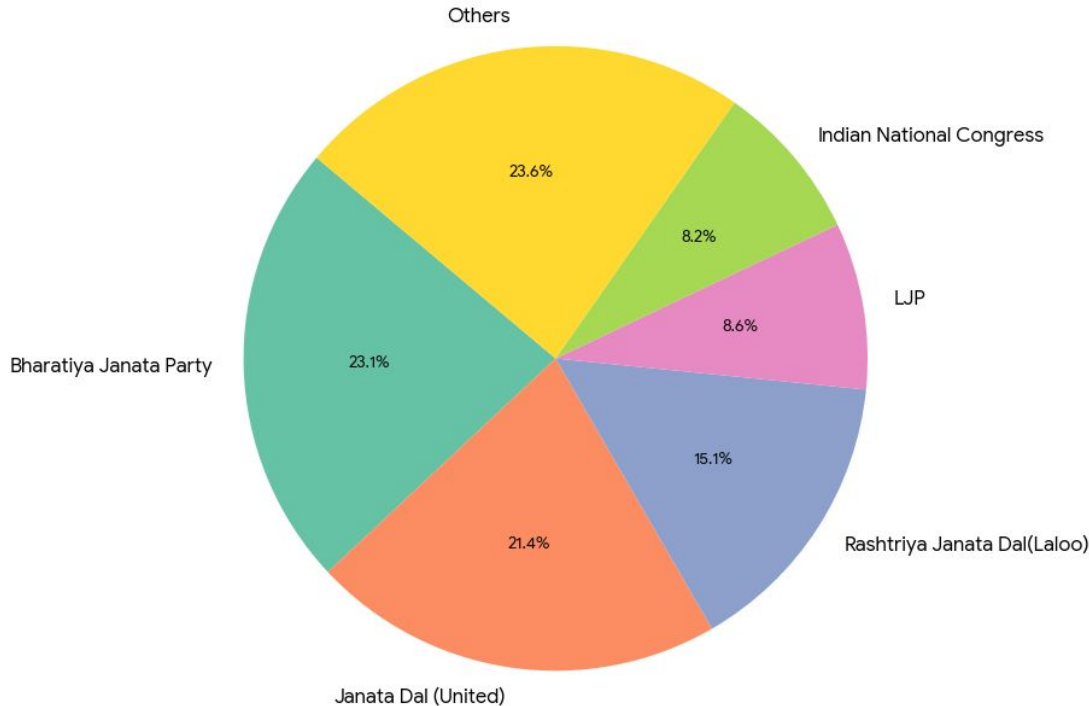
Inference :

- Voting in the state shows **dominance of major parties**.
- The vote share distribution reflects **clear electoral trends** and voter alignment.

Party-wise Vote Share – Bihar (Lok Sabha 2019)

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Vote Share by Party: Bihar (2019)



Objective :

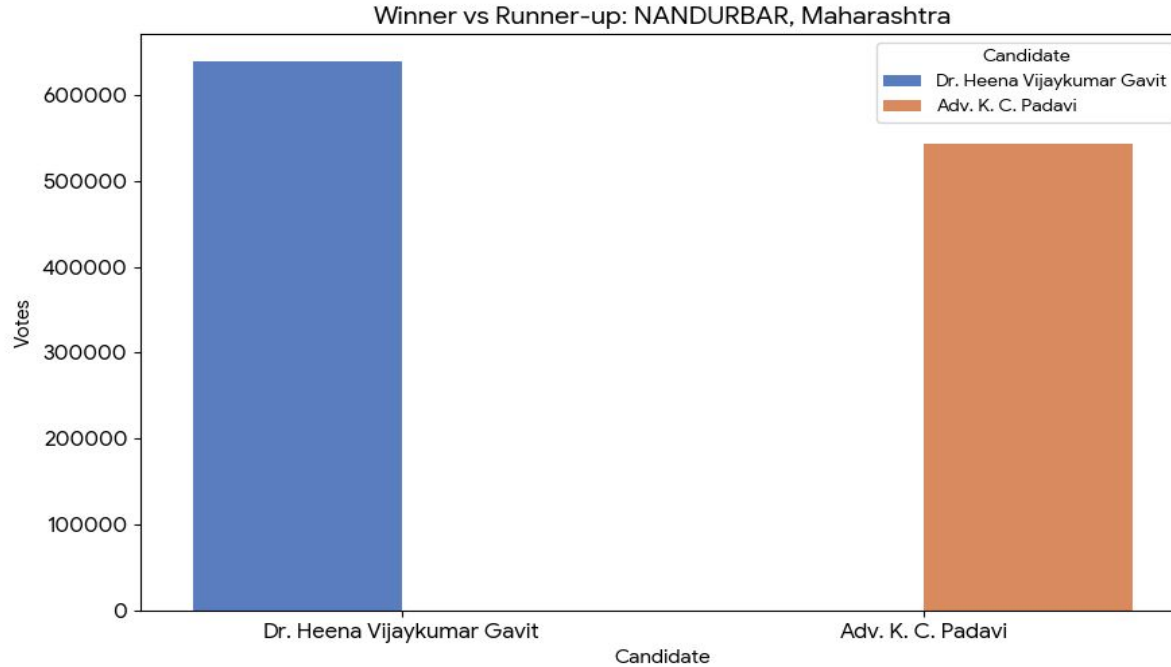
To understand the **distribution of votes among political parties** in ***Bihar*** during the **2019 Lok Sabha election**.

Observation :

- The leading party secured the **highest share of votes** in the state.
- Major national and regional parties together account for a **significant portion of total votes**.
- Smaller parties contribute a **limited share**, indicating concentrated voter preference.

Winner vs Runner-up: Maharashtra

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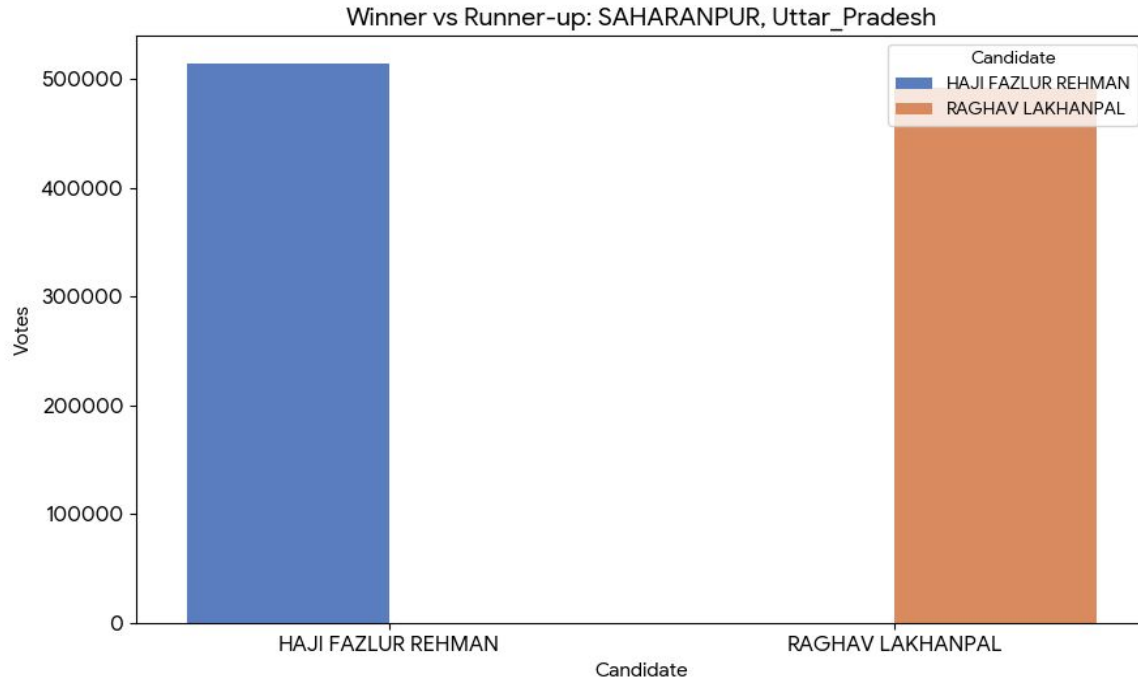
Observation :

The winning candidate received more votes than the runner-up.

The vote difference reflects a clear winning margin.

Winner vs Runner-up : Uttar Pradesh

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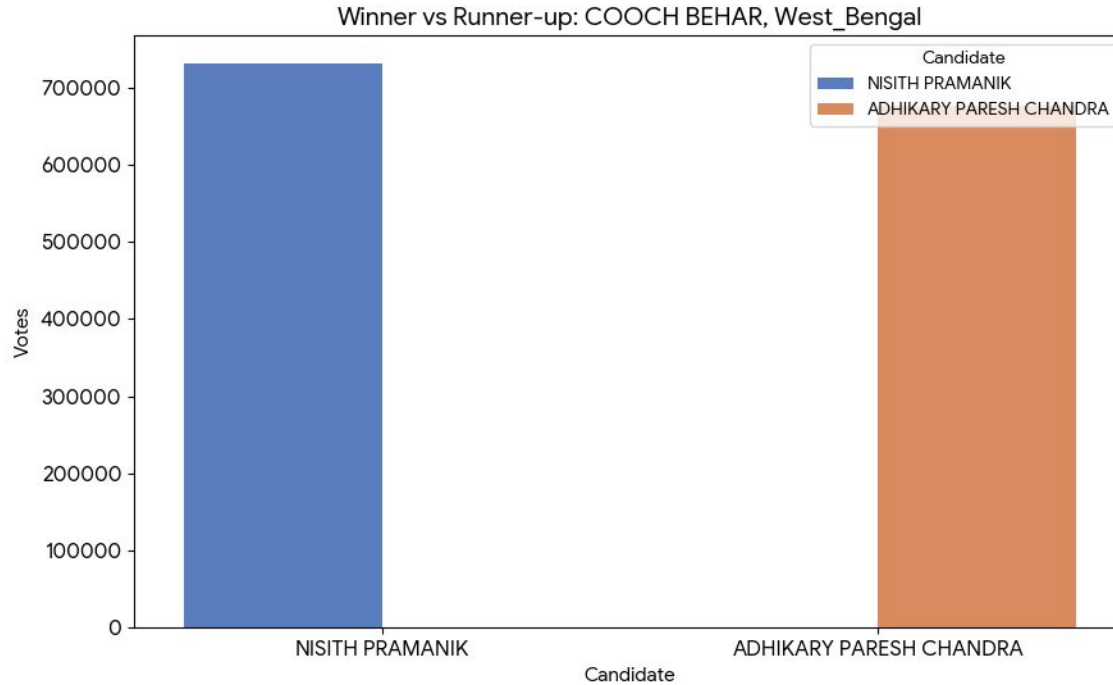


Observation :

- The winning candidate, **Haji Fazlur Rehman**, secured a **higher number of votes** compared to the runner-up.
- The bar graph highlights the **performance gap** between the two leading candidates.
- This reflects the **strength of voter support** for the winning candidate.

Winner vs Runner- West Bengal

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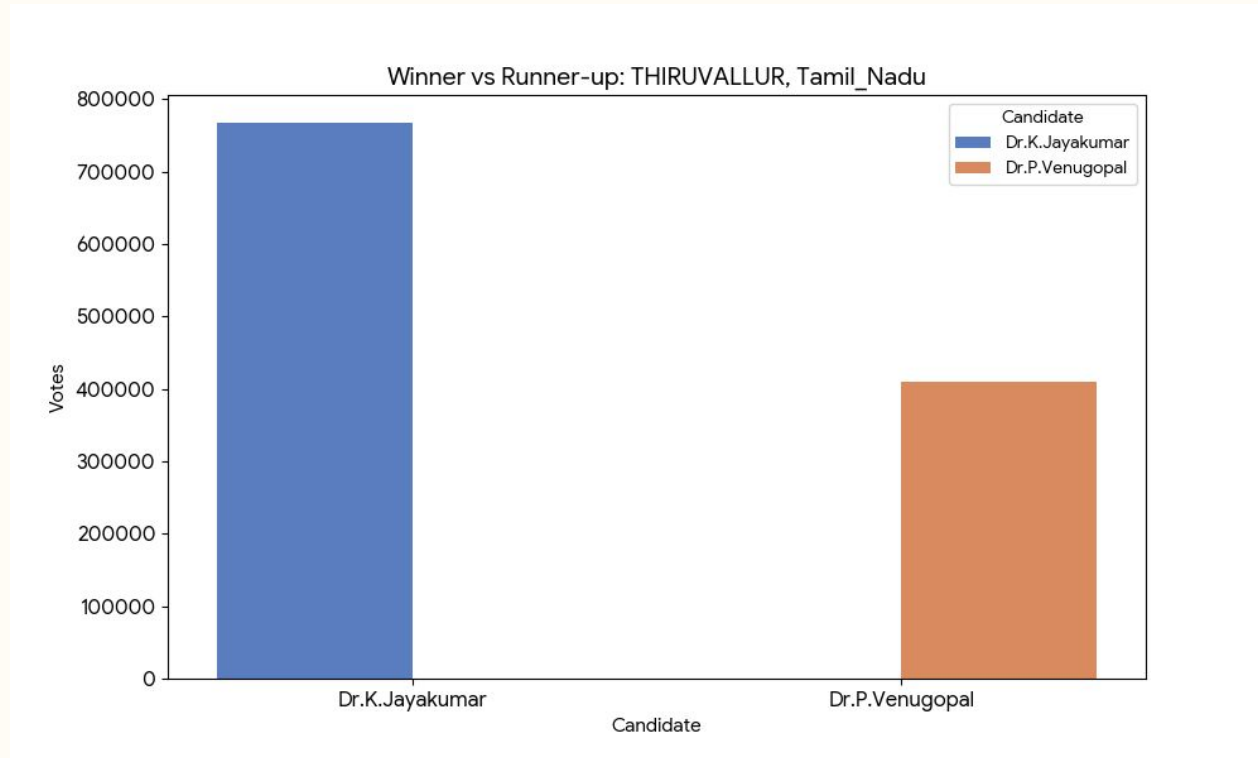


Observation :

- The difference in votes indicates a **clear margin of victory**.
- The result reflects **strong voter support** for the winning candidate.

Winner vs Runner - Tamil Nadu

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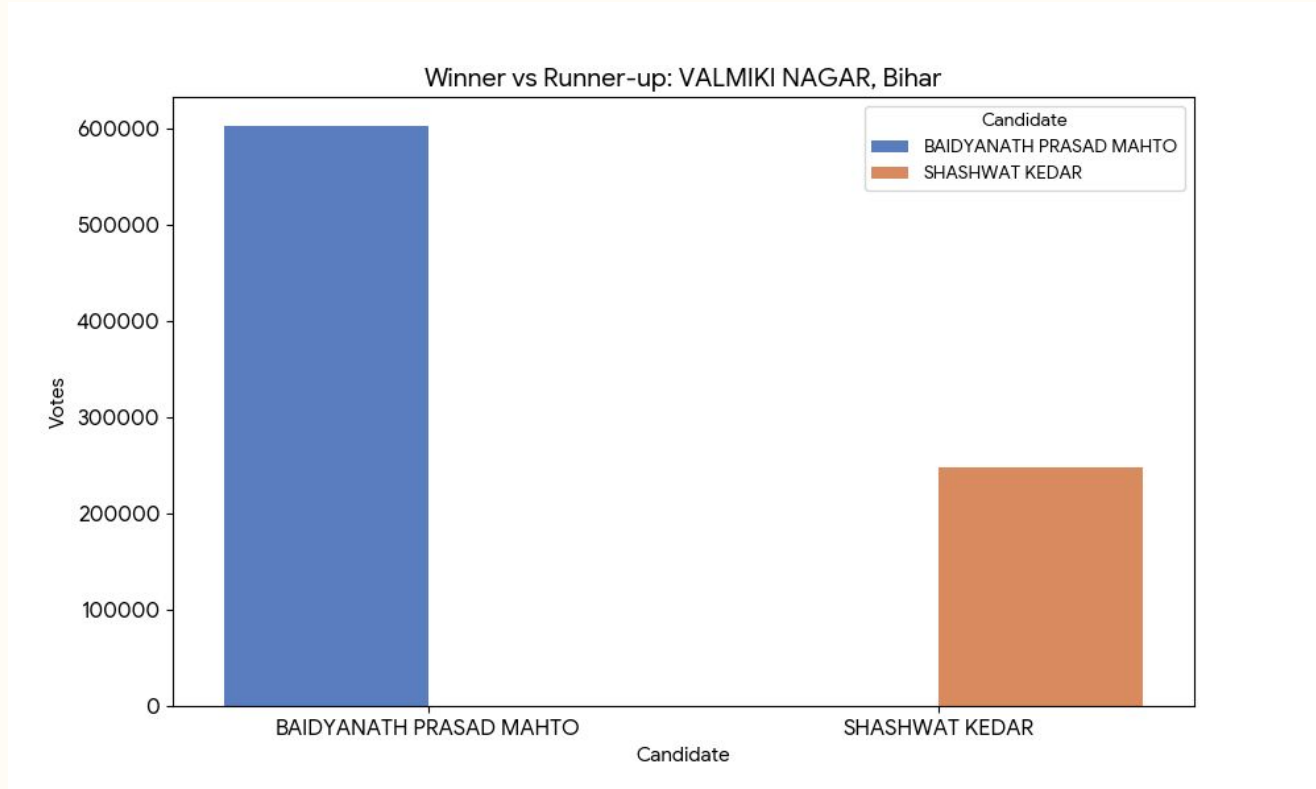


Observation :

- The winning candidate received a higher number of votes than the runner-up.
- A clear difference in votes is visible in the bar graph.

Winner vs Runner - Bihar

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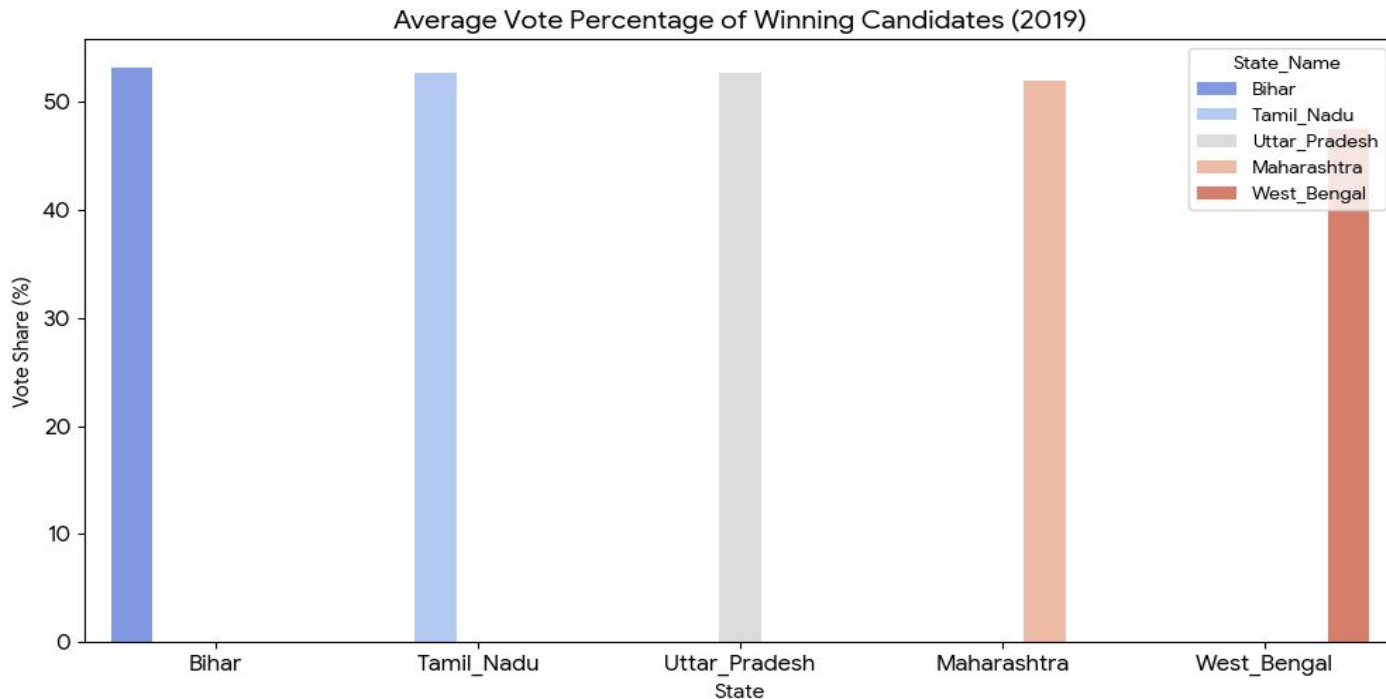


Observation :

The bar graph shows that the winning candidate received **significantly more votes**, indicating a **clear electoral margin and strong voter support**.

Comparison of Voting Percentage of Winning Candidates (5 States)

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To compare the **voting percentage of winning candidates** across the **five selected states** in the Lok Sabha Election 2019.

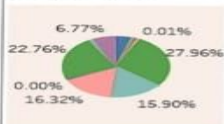
The column graph highlights **variations in winning vote percentages across states**, reflecting differences in electoral competition and regional voting patterns.

Tableau Dashboard – Lok Sabha Election Analysis (2019)

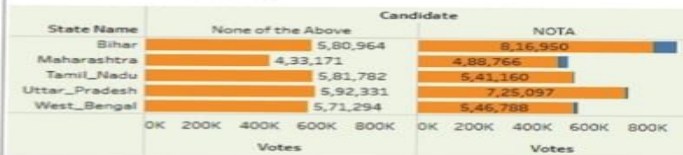
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Lok Sabha 2019 Election Analysis – Selected 5 States

Party Vote % – State wise



Winner vs Runner-up



- State Name
- ☐ (All)
 - ☐ Andaman_&N
 - ☐ Andhra_Prade
 - ☐ Arunachal_Pre
 - ☐ Assam
 - ☒ Bihar
 - ☐ Chandigarh
 - ☐ Chhattisgarh
 - ☐ Dadra & Nagar
 - ☐ Dadra_&_Naga
 - ☐ Daman_&_Diu
 - ☐ Delhi
 - ☐ Goa
 - ☐ Goa_&_Daman_
 - ☐ Gujarat
 - ☐ Haryana

- Party Name
- ☒ Null
 - ☐ Aam Adhikar M
 - ☐ Aam Janta Par
 - ☐ Aapki Apni Par
 - ☐ Agila India Mai
 - ☐ Aihra National
 - ☐ Akhil Bharatiya
 - ☐ Akhil Bharatiya
 - ☐ Akhil Bhartiya
 - ☐ All India Majlis
 - ☐ Ambedkar Nati
 - ☐ Ambedkar Sam
 - ☐ Ambedkarist R
 - ☐ Amra Bangalee
 - ☐ Anti Corruption

- Party
- ☒ Null
 - ☒ NOTA

Winning Candidates – Column Graph



Dashboard Design & Methodology

The dashboard was created in Tableau using interactive features such as Pages Shelf for state-wise analysis, filters to restrict data to selected states, and color encoding to represent party-wise performance. Multiple charts were combined into a single dashboard for comparative analysis.

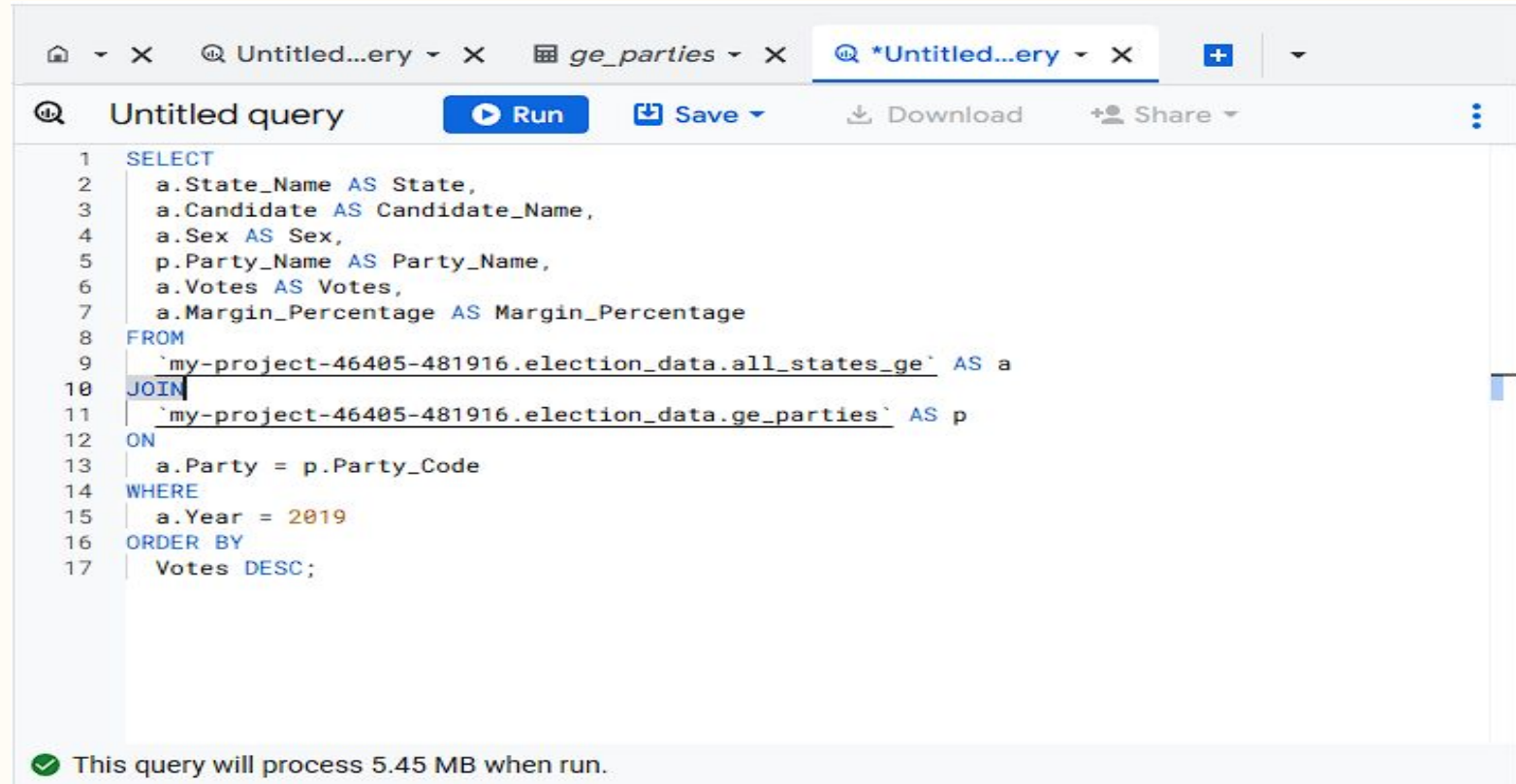
Dashboard Shows :

- i. Party-wise vote share
- ii. Winner vs runner-up comparison
- iii. Winning candidates performance

Interactive Dashboard Link :

<https://public.tableau.com/app/profile/sakshi.wagh4695/viz/LokSabha2019ElectionAnalysis5States/LokSabha2019#1>

- SQL JOIN query :

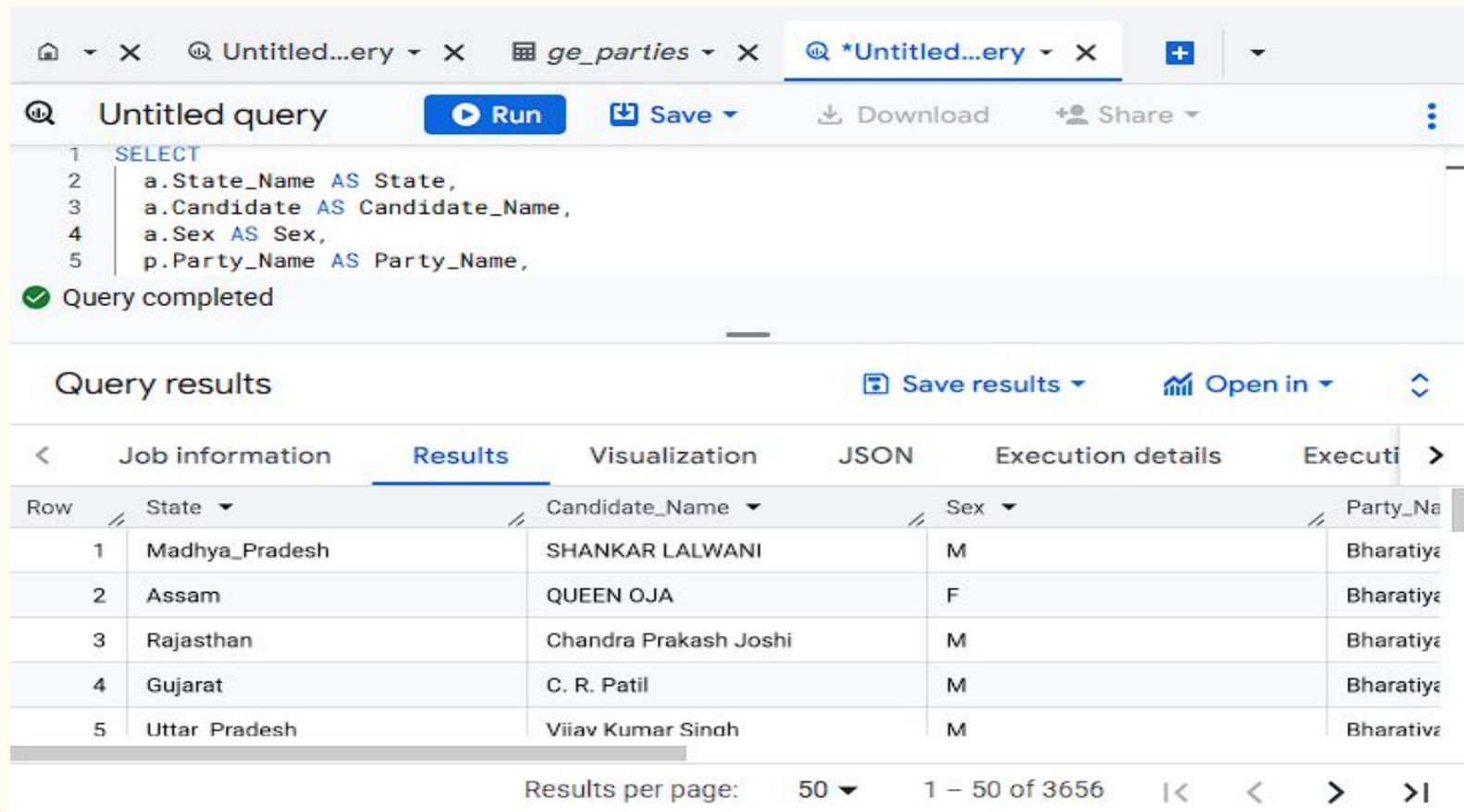


The screenshot displays the Google Cloud BigQuery console interface. At the top, there are tabs for 'Untitled...ery', 'ge_parties', and '*Untitled...ery'. The active tab is '*Untitled...ery'. Below the tabs, the query editor shows an SQL query with line numbers 1 through 17. The query is a JOIN query that selects columns from two tables: 'election_data.all_states_ge' (aliased as 'a') and 'election_data.ge_parties' (aliased as 'p'). The query filters for the year 2019 and orders the results by the number of votes in descending order. The query is as follows:

```
1 SELECT
2   a.State_Name AS State,
3   a.Candidate AS Candidate_Name,
4   a.Sex AS Sex,
5   p.Party_Name AS Party_Name,
6   a.Votes AS Votes,
7   a.Margin_Percentage AS Margin_Percentage
8 FROM
9   `my-project-46405-481916.election_data.all_states_ge` AS a
10 JOIN
11   `my-project-46405-481916.election_data.ge_parties` AS p
12 ON
13   a.Party = p.Party_Code
14 WHERE
15   a.Year = 2019
16 ORDER BY
17   Votes DESC;
```

Below the query editor, a status message indicates: "This query will process 5.45 MB when run."

Output table :



The screenshot shows a SQL query editor interface. At the top, there are tabs for 'Untitled...ery', 'ge_parties', and '*Untitled...ery'. The active tab is '*Untitled...ery'. Below the tabs, there is a search bar with 'Untitled query' and buttons for 'Run', 'Save', 'Download', and 'Share'. The query text is as follows:

```
1 SELECT
2   a.State_Name AS State,
3   a.Candidate AS Candidate_Name,
4   a.Sex AS Sex,
5   p.Party_Name AS Party_Name,
```

Below the query, a green checkmark indicates 'Query completed'. The 'Query results' section shows a table with 5 rows and 5 columns. The table has tabs for 'Job information', 'Results', 'Visualization', 'JSON', 'Execution details', and 'Execution'. The 'Results' tab is selected. The table data is as follows:

Row	State	Candidate_Name	Sex	Party_Name
1	Madhya_Pradesh	SHANKAR LALWANI	M	Bharatiya
2	Assam	QUEEN OJA	F	Bharatiya
3	Rajasthan	Chandra Prakash Joshi	M	Bharatiya
4	Gujarat	C. R. Patil	M	Bharatiya
5	Uttar Pradesh	Vijay Kumar Singh	M	Bharatiya

At the bottom, there is a pagination bar showing 'Results per page: 50', '1 - 50 of 3656', and navigation arrows.

Explanation :

- The election results table (**all_states_ge**) was joined with the party master table (**ge_parties**) using Party_Code as the key
- SQL WHERE clause filtered data for the year 2019, and ORDER BY arranged candidates in descending order of votes. This enables analysis of party performance and candidate rankings.”.

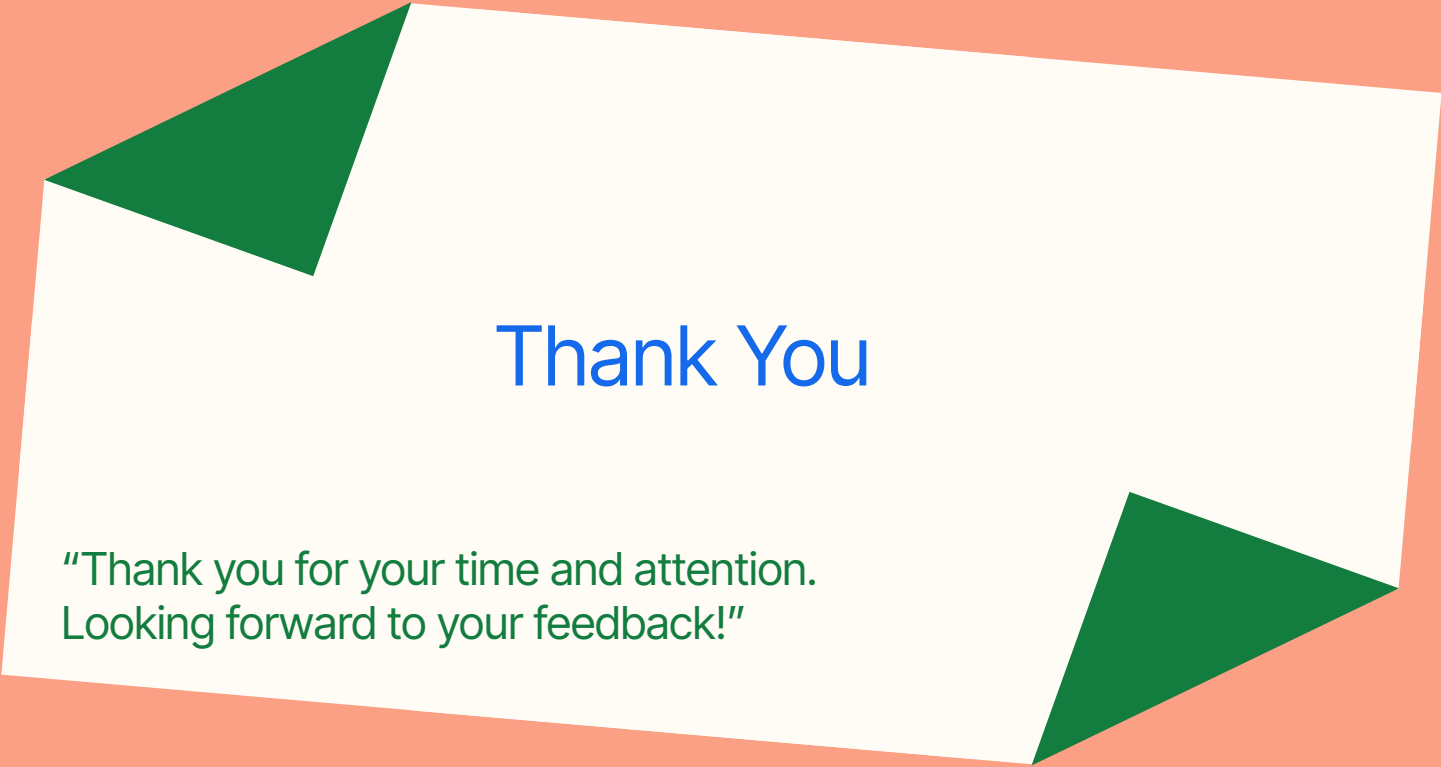
As shown in output we can clearly observe that the query run successfully and the result is correctly formatted

Conclusion

This project demonstrates how election data can be analyzed effectively using Google Sheet , Tableau and BigQuery.

Visualizations of party-wise vote share, winner vs runner-up comparisons, and winning candidate performance provide clear insights across five selected states for the 2019 Lok Sabha elections.

SQL queries and Tableau dashboards together enabled a comprehensive understanding of candidate and party performance.



Thank You

"Thank you for your time and attention.
Looking forward to your feedback!"